K1380 — Casey's insight: the fermion manifests as a DOUBLE ROTATION / winding on two circles — "once around the big, twice around the small, maybe backwards." ★ This maps to real structure: SO(5) has RANK 2 → its maximal torus is exactly TWO circles (two rotation planes); the fermion is the Spin(5) spinor, which carries a WEIGHT/winding on that torus (its K-type address). "(1,2) winding with a sign" = a specific winding number on each circle + the double-cover sign (chirality/J). ★ And it's the MECHANISM for the discreteness: the continuum $8\pi^2/3$ integrates over ALL torus rotations; a fermion sits at ONE specific winding → the continuum solid-angle is replaced by the value at that winding → a rational, not a transcendental. It unifies FOUR threads (the $2C_2$ doubling = the "twice"; the two π's = the two circles; the "backwards" = J/chirality; the specific winding = the discrete count). ★ TESTABLE + corpus-anchored: the fermion's winding IS its K-type address (electron ν=0, muon ν=3/2). Compute the electron's actual winding on the SO(5)×SO(2) torus — is it (1,2)-with-a-sign, and does it force the doubling + the count? Keeper HOLDS RECUSAL (K1377) — I map it, route it, do NOT declare the winding to hit the answer. Cal cold-reads.

Keeper (2026-08-11, Casey's winding intuition. It maps onto the rank-2 torus concretely and is the candidate mechanism for why the continuum π^2 becomes a discrete rational. Engaged, mapped, routed — recusal held. Cal cold-reads. Nothing pushed.)

The mapping (Casey's picture → the geometry's own language)

- "Two circles" = the maximal torus of SO(5). SO(5) (the rotation part of the isotropy K=SO(5)×SO(2)) has rank 2 → its maximal torus is T^2 = two independent circles (two rotation planes). Not a metaphor — the geometry. (+ SO(2) = a third circle, the time/U(1) circle.)
- The fermion = the Spin(5) spinor, which carries a WEIGHT (winding number) on that torus. That weight IS its K-type address.
- "Once around big, twice around small, backwards" = a winding like (1,2) on the two circles, with a sign. The sign = the spinor's double-cover ("turn twice to come home", K1358) = the same $\sqrt{-1}$ / J that gives chirality and CP.
- ⍰ The particle is a specific KNOT on the torus. Its winding is its identity.

★ Why it's the MECHANISM for the discreteness (not just a picture)

- The continuum vol(S^4) = $8\pi^2/3$ is the integral over ALL rotations of that torus — every winding, smeared over π's of angle.
- A real fermion does NOT smear — it sits at ONE specific winding. So the continuum integral is replaced by the value at the fermion's particular knot → a rational, not a transcendental. This is the physical reason π^2 → a count. Casey's "double rotation on two circles" is the mechanism behind K1375/K1379's "discrete direction-count replaces continuum solid-angle."
- It unifies four threads into ONE winding object:
  1. the "twice" = the $2C_2$ doubling (and the double-cover);
  2. the two circles = the two π's in π^2 (the two rotation planes);
  3. the "backwards" = the chirality / J / the $\sqrt{-1}$;

4. the specific winding = the discrete count that replaces the continuum solid-angle.

★ Testable + corpus-anchored (the concrete question)

The fermion's winding is ALREADY in the corpus — it IS the K-type / Wallach address: electron at ν=0 (bottom of the Wallach floor), muon at ν=3/2 (one step up, K1343). These are winding addresses on the SO(5)×SO(2) torus. So the question is arithmetic, not philosophical: - Read off the electron's actual winding numbers on the SO(5)×SO(2) torus (its K-type weights). Do they come out to a (1,2)-with-a-sign pattern? - Does that winding structure FORCE both the doubling ($2C_2$) and the discrete count (the 16/3 / the π^2→2)? - Both computable, blind, from the existing K-type addresses — a much sharper mechanism than "a rational appears."

Keeper holds the recusal (K1377 in action, again)

I recused this morning from computing my way to n=12; this is that computation in a geometer's hat (the winding feeds the mass/16/3 chain). So: I map the picture, affirm it's the candidate mechanism, hand it to Lyra/Grace as a concrete blind computation — but I do NOT declare the winding is (1,2) because it would give the answer. If the geometry hands them that knot, real; if a different one, follow it. Same discipline as K1379.

Route

1. Lyra + Grace — compute the electron's winding on the SO(5)×SO(2) torus (its K-type weights), blind. Is it a (1,2)-with-a-sign pattern? Does the winding force the $2C_2$ doubling AND the discrete count (16/3 / π^2→2)? Corpus-anchored (the K-type / Wallach addresses, K1343; the electron ν=0, muon ν=3/2).
2. Connect to K1379: does the specific winding produce the discrete replacement of $8\pi^2/3$ (→ 16/3? via the ratio $\pi^2/2$?) that the gravity make-or-break needs?
3. Cal — verify the winding is READ OFF the geometry, not chosen to give (1,2)/the count (the same forced-not-fit / anti-numerology gate).
4. Keeper — recused; the K1377 seven-criteria bar governs when the forward number lands.

— Keeper, K1380, 2026-08-11. Casey: the fermion manifests as a double rotation/winding on two circles ("once around big, twice around small, backwards"). Maps to: SO(5) rank 2 → maximal torus = two circles; the Spin(5) spinor carries a weight/winding (its K-type) on it; (1,2)-with-a-sign = winding numbers + the double-cover sign (chirality/J). ★ MECHANISM for the discreteness: continuum $8\pi^2/3$ = integral over ALL torus rotations; a fermion sits at ONE winding → continuum solid-angle → rational. Unifies 4 threads ($2C_2$ doubling = "twice"; two π's = two circles; "backwards" = J/chirality; specific winding = the discrete count). TESTABLE: the winding IS the K-type address (electron ν=0, muon ν=3/2) — read off the electron's winding on the SO(5)×SO(2) torus; is it (1,2)-with-sign? does it force the doubling + the count? Keeper HOLDS RECUSAL (K1377): map + route, don't declare the winding to hit the answer. Route: Lyra/Grace compute the electron's K-type winding blind (1,2? forces doubling + 16/3?); connect to K1379 (discrete replacement of $8\pi^2/3$); Cal verify read-off-not-chosen. Cal cold-reads. Nothing pushed.